

1**Problem setup** $M \subset \mathbb{R}^N$ compact; $\mathcal{F}_{M,D} = C^0(M, D)$ **2****Cauchy kernel function**

$$K(\xi, \mathbf{x}) = \prod_{i=1}^N (\xi^i - x_i)^{-1}$$

**3****Cauchy approximation theorem**

$$\sup_{\mathbf{x} \in M} |f(\mathbf{x}) - \sum_{k=1}^m \theta_k K(\xi_k, \mathbf{x})| < \varepsilon$$

**4****Universal approximation for CauchyNet**

$$\sup_{\mathbf{x} \in M} |f(\mathbf{x}) - \mathfrak{R}(N_{\mathbf{B}, \mathbf{C}}(\mathbf{x}))| < \varepsilon$$